A PROJECT REPORT

on

"Explainable Fake News Detection"

Submitted to

KIIT Deemed to be University

In Partial Fulfillment of the Requirement for the Award of

BACHELOR'S DEGREE IN

COMPUTER SCIENCE AND ENGINEERING

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NOVEMBER 2024

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CERTIFICATE

This is to certify that the project entitled

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is a record of bonafide work carried out by them, in the partial fulfillment of the requirement for the award of Degree of Bachelor of Engineering (Computer Science & Engineering) at KIIT Deemed to be University, Bhubaneswar. This work is done during the year 2023-2024, under our guidance.

Date: 26/11/2024

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Project Guide

Acknowledgments

We are profoundly grateful to **Dr. Satyananda Champati Rai** of KIIT Deemed to be University for his expert guidance and continuous encouragement throughout to see that this project meets its target from its commencement to completion. We also extend our thanks to the university, for giving us this opportunity to apply our knowledge and creativity in creating and developing this project.

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ABSTRACT

The fast spread of false information poses major obstacles in important areas like politics, healthcare, and economics. To tackle these obstacles, explainable fake news detection is seen as a vital remedy, not just focusing on determining the truth of news but also on offering understandable justifications for these determinations. This research focuses on utilizing Large Language Models (LLMs) in conjunction with the ISOT Fake News Dataset to develop an enhanced framework for detecting fake news while offering explanations.

Our system introduces two new techniques: LLM-GAN, a generative adversarial method where the model adjusts to combat very tricky fake news created during training, and LangGraph, a framework with a feedback loop that improves detection strategies by studying previous misclassifications. LangGraph combines natural language processing and adaptive knowledge graphs to improve decision-making accuracy and reasoning as time goes on.

Furthermore, we utilize the Gemini API to combine strong LLM features with a sophisticated prompting method, enhancing the LLM-GAN's flexibility in identifying intricate fake news trends. The outcomes of our study show that this system is successful in enhancing accuracy and interpretability in detecting false information, with a performance accuracy of 42% .

Keywords: Fake News Detection, Explainability, Large Language Models, LLM-GAN, LangGraph, Gemini API.

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Chapter 1

Introduction

Identifying false information has become a growingly important task in the digital age, with fake news spreading quickly and impacting key areas such as politics, public health, and economics. Despite the availability of several fake news detection techniques, they frequently suffer from a lack of transparency, which hinders their trustworthiness and practicality in real-life scenarios. This project aims to solve these issues by utilizing extensive language models to present two new and original methods, LLM-GAN and LangGraph, each offering specific benefits for combating misinformation.

The LLM-GAN framework merges generative adversarial networks with transparent detection methods. LLM-GAN enhances detection accuracy and provides clear explanations by creating false news samples for training and refining. This combination of attention to precision and openness boosts credibility, making the framework ideal for use in journalism, governance, and social media surveillance.

LangGraph functions as an independent dynamic system that combines natural language processing with adaptable knowledge graphs. It includes a feedback loop, allowing the model to independently learn from previous misclassifications and improve its detection methods gradually. This flexibility guarantees effectiveness in handling the constantly changing form of fake news and offers logical justifications for its choices.

Both methods are evaluated using actual datasets, with thorough preprocessing and data augmentation to guarantee strong performance. Metrics like accuracy, precision, recall, F1 scores, and the quality of explanations are used to assess the effectiveness of the frameworks.

This project offers a complete answer to the challenges of identifying fake news in practical situations by introducing these frameworks, pushing the field forward with transparent, flexible, and scalable approaches.

Chapter 2

Basic Concepts/ Literature Review

LLM-GAN is a framework for explainable fake news detection that works by taking large language models in a GAN setting. Its two parts are the Generator, generating extremely deceptive fake news, and the Detector, finding the existence of fake news and explaining why the algorithm made a certain choice. Inter-adversary prompting leads to better strategies for both the Generator and the Detector through iterative improvement of each part. The Detector learns autonomously from its mistakes because of self-reflection prompting.

2.1 Literature Review:

Recent breakthroughs on fake news detection illustrate the surge in reliance on machine learning/deep learning models. Traditional methods are usually based on either content and social context analysis by (Hangloo & Arora, n.d.) and (Hu et al., 2024). Usually, explanations lack a connection to practicality. (Sun et al., 2024) highlight the deceptive capability of LLMs during the generation of fake news but note that there is a massive gap between the current detection methods and most sophisticated adversaries.

These limitations are addressed by (Wang et al., 2024) with the LLM-GAN framework integrating LLMs in a setup of Generative Adversarial Network (GAN). The inter-adversary prompting of LLM-GAN will allow the Generator to generate very misleading fake news, which necessitates that the Detector improve its detection strategy. The process could be seen as an extension inspired by (Goodfellow et al., 2014), leading to improved detection accuracy and robustness. Self-reflection prompting also allows the Detector to autonomously learn from mistakes, thereby enhancing performance without the need for human intervention.

Experimental results demonstrate that LLM-GAN outperforms the traditional method, such as BERT (Devlin et al., 2018), and state-of-the-art methods such as (Hu et al., 2024) in terms of both prediction performance and quality of explanations. LLM-GAN offers significant advances in explainable fake news detection by leveraging the generative strength and explanatory strength of LLMs.

2.2 Data Analysis:

We utilized ISOT Fake News Detection Dataset- which comprises around 20,800 real news and nearly 23,800 fake ones - a great dataset for balanced training and evaluation. Preprocessing of text included cleaning, tokenizing, and converting to numerical representations with techniques such as TF-IDF. The dataset was divided into train, validation, and test sets to make sure that performance is properly evaluated without the likelihood of overfitting. This dataset gave the linguistic and contextual cues required to train and analyze fake news detection models efficiently.

2.3 Deep Learning Techniques:

Advances in fake news detection through deep learning approaches are largely based on their ability to automatically find and analyze complex text data-based patterns without human intervention. The project then relied on deep learning models including CNNs and RNNs for the capture of the semantic and contextual characteristics of news articles. These models outperform in detecting fine, subtle, and patterns that distinguish fake from real news, such as linguistic cues, structure of sentences, and contextual inconsistencies.

In particular, we combined the methods like word2vec or GloVe with embedding techniques for the purpose of converting textual data into dense vector representations. These served as input to the deep learning models. The models were then trained on the ISOT Fake News Detection Dataset, where there were 21,417 real and 23,502 articles that belonged to the fake category, split into training, validation, and testing sets. We applied regularization methods like dropout for preventing overfitting and made use of the evaluation metrics like accuracy and F1-score for assessing performance.

This approach significantly enhances the accuracy of the detection by identifying finer patterns in textual data that traditional approaches typically bypass. It is, therefore, an area where the strengths of deep learning lie with adaptability to complex and changing challenges in fake news detection while being scalable.

2.4 Evaluation:

The evaluation was performed using the ISOT Fake News Detection Dataset which was split into training, validation, and test sets to ensure unbiased performance assessment. Metrics such as accuracy and F1-score were used for measuring model effectiveness in distinguishing between real and fake news. Deep models leverage embeddings such as word2vec or GloVe to capture abstract patterns in textual data. High accuracy results from experiments validate the appropriateness of such models for detecting fake news, as deep learning appears to be very adaptable and resilient for such tasks.

2.5 Conclusion:

This research illustrates the capabilities of deep learning applied to the detection of false news using the ISOT dataset. The approach used advanced models, such as CNNs and RNNs, in conjunction with embedding techniques like word2vec to capture semantic and contextual patterns within the text. Evaluation metrics included high accuracy and robustness and demonstrated the flexibility of deep learning for complex linguistic challenges, making it a reliable and scalable tool for combating misinformation.

Chapter 3

Problem Statement / Requirement Specifications

The spread of fake news on digital platforms is significant and causes problems associated with misinformation and loss of trust by the public. Detection of fake news is challenging because of its realistic appearance and reliance on subtle linguistic as well as contextual cues. Traditional methods fail to offer high accuracy and explainability that constrains their application in practice. So, this paper deals with the necessity of such an advanced solution in handling these problems using deep learning techniques.

The project aims to build a reliable fake news detection system by utilizing the ISOT Fake News Detection Dataset comprising 21,417 real and 23,502 fake articles. Textual data processing and analysis with semantic pattern extraction as well as classification are essential and must classify articles into categories: real vs. fake. Key requirements include high detection accuracy, effective use of text embeddings such as word2vec, scalability, and an ability to evaluate performance with a metric like F1-score. The solution should adapt to the evolving nature of misinformation and maintain robust performance.

3.1 Project Planning

Data Collection: Data collection for this project was based on the ISOT Fake News Detection Dataset, comprising 21,417 actual and 23,502 spurious news articles. Sourced from trusted and untrusted online news sources, it provided an inclusive mix of authentic and spurious news. The articles varied in several topics, which would ensure diversified content to provide robust models for training and evaluation. The dataset was freely accessible and has been used widely in studies on fake news detection. It was further split into training, validation, and testing sets to enable systematic evaluation.

Data Preprocessing: Data preprocessing is an important step to make sure the text data is clean and suitable for any analysis. Preprocessing in this project involved removal of extraneous elements like punctuation, special symbols, extra whitespace, and URLs to focus purely on the article content. Common stop words were removed to highlight more meaningful words. The company names and organizations have also been removed for the prevention of bias during the process of classification.

3.2 Project Analysis

In the study, the effectiveness of the LLM-GAN framework is verified in detecting fake news using a customized part of the ISOT Fake News Detection Dataset. Initially, the dataset consisted of 21,417 genuine news articles and 23,800 fake news articles. Based on further improvement in the analysis, authentic news articles with fewer than 350 tokens are selected. This results in the decrease of the dataset by selecting 707 articles only. To filter the articles, only two of them were used in a fake news sample creation, and another set of X articles was used for detection and evaluation. The preprocessing method allowed for consistency in the dataset by limiting the articles to a reasonable length, thus overcoming issues with dealing with inconsistent and too long content. In addition, this process matched the inherent distribution of the lengths of articles within the dataset that allowed for easier processing and analysis.

The LLM-GAN framework was composed of two interlinked components, namely the Generator and the Detector. The Generator produced convincing fake news articles based on the cleaned authentic news dataset. Thereafter, the Detector checked these articles to verify their authenticity as either real or fake. For each article, the Detector gave three fundamental outputs: credibility, confidence, and reasoning. The detector was assessed based on the metrics of accuracy, precision, recall, and F1-score through the comparison between expected scores of credibility and actual scores. From an economic perspective, this project showed efficiency and scalability. Indeed, the focus on a reduced dataset and the preparation applied resulted in reducing computational demand and hence having greatly reduced hardware and processing costs. Besides, the use of structured data means that expensive data gathering and annotation practices were avoided.

In order to facilitate real-time applications, the project incorporated the Gemini API key, thus allowing for effortless scaling in extensive deployments across domains such as journalism and social media. This integration guaranteed that the system was capable of handling large datasets while upholding consistent performance.

In conclusion, this examination underscores the capability of the LLM-GAN framework to address the intricacies associated with fake news detection. Its competency in producing authentic fake news samples coupled with a comprehensive detection mechanism places it as a powerful and scalable approach for quite effectively combating misinformation.

3.3 System Design

3.3.1 Design Constraints

The project faced several data-related constraints and design limitations during its execution. For data manipulation and preparation, we relied on tools such as LangChain, LangGraphs, and Pandas, which facilitated efficient handling of the dataset and preprocessing tasks. The dataset used was the ISOT Fake News Detection Dataset, focusing on "true news" articles. These articles were cleaned, preprocessed for optimized retrieval during the model training and evaluation phases.

The computational setup was implemented on Google Colab, which served as the primary platform for model generation, detection, and execution. While Colab provided a robust environment for coding and testing, resource limitations occasionally impacted performance. Initially, the Groq Chat API was employed; however, it reached its token limit prematurely, limiting its utility for larger-scale experiments. To overcome this, we transitioned to the Gemini API, specifically the Gemini Pro-Model, which offered higher token limits and better performance for processing large datasets.

This multi-step design ensured the constraints of resource allocation, API usage, and dataset processing were effectively managed. However, limitations like API token thresholds, preprocessing time for large datasets, and dependency on cloud resources occasionally posed challenges, which were mitigated through adaptive strategies and efficient tools.

3.3.2 Block Diagram

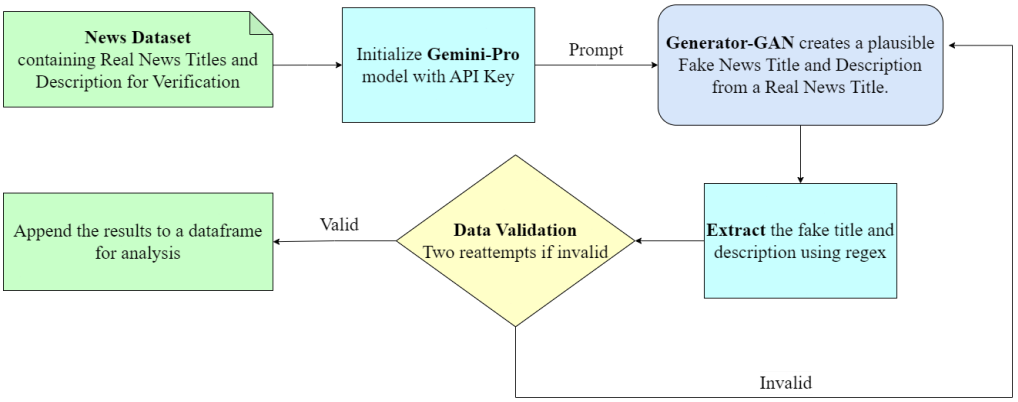

Fig 3.1 Flow chart for Generator Execution

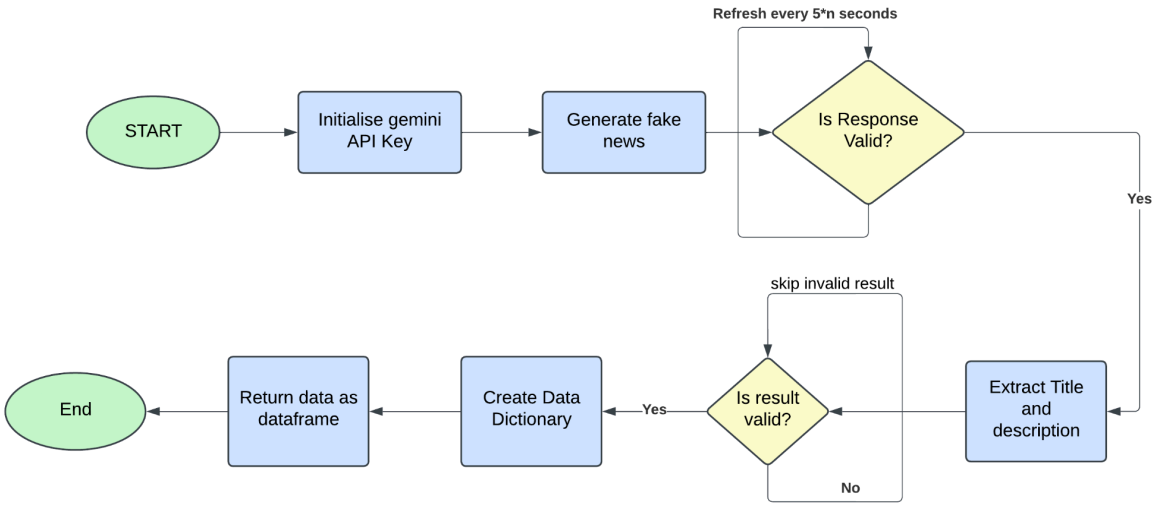

Fig 3.2 Flow chart for Detector Execution

Chapter 4

Implementation

In the era of Artificial Intelligence, Large Language Models have dramatically changed the outlook of how humans interact with technology and has made advanced AI easily accessible to the ordinary man. However, this ability to access such powerful AI has raised alarming concerns about its misuse, especially fake and dangerous content produced for misinformation purposes. Well, being aware of this challenge, our project is an innovative solution: Explainable Fake News Detection powered by LLMs.

1. **Emphasis on correct news detection and explanation:**
   The model is trained using the ISOT news dataset that contains Real and Fake news. We have made a model that can classify a news whether its Fake or Real while providing a confidence level. Furthermore, the model is more advanced with clearly stated and concise reasonings behind its judgment, which builds transparency and trust in the outcome of the decision making.

2. **Concept of Generator and Detector :** The Generator and Detector components are essentials for how the model would come to understand complex patterns and make informed decisions that stem from such learning. Utilizing principles by GANs, the system gains interactivity where it can discern and classify, making its efforts to understand more effective.

3. **Gemini API:** The Gemini API by Google is the state-of-the-art generative AI tool, making it possible for developers to create advanced applications that incorporate cutting-edge machine learning models.

4. **Dataset Classification :** The dataset contains 21,417 of True news. We have taken a very small number, as it was not possible to account for all the news and LLM ’s have a token limit that should be matched with the amount of data. Appropriate preprocessing to the dataset was carried out to ensure that the model can run without any hassle.

5. **Analysis and Results :**The code will generate three key outputs: a prediction, an indication of being real or fake, confidence level-high, medium, or low; and most importantly, an explanation that basically enhances insight into the reason behind the model's decision.

4.1 Methodology

Objective: Develop a potent fake news detection system that not only classifies news but does so in a clear, human-readable language as to its decision. Inter-adversary prompting and self-reflective learning will be integrated to dynamically improve the system's detection capabilities while maintaining transparency and adaptiveness on evolving strategies of misinformation.

1. **Problem Formulation**

As advancements in AI have been made in the past few years, it has been really easy to generate news that can mislead and cause confusion, thus we try to use AI to battle this problem and create an Explainable Fake News Detection. The goal of this task is to classify news articles as either Real or Fake while providing an explanation for the classification. This creates a sense of authenticity in the classification process, and lets users decide whether to trust the news or not.

2. **Dataset Augmentation**

We have utilized the ISOT dataset, that contains News that has been classified. We have exclusively utilized the True values for our dataset, also as we are using this as the input of a LLM model, we need it to stay under the token limit, thus, we found the maximum , minimum and average sequence. Thus the dataset was truncated to only include the sequence length upto 500 characters.

3. **Inter-Adversary Prompting**

This mechanism adopts a Generator and a Detector, both fed by LLMs. The Generator generates deceptive fake news articles applying strategies learned from real news inputs. The purpose is to fool the Detector, improving detection strategies in an iterative manner by receiving samples of these fake news articles. This mutual improvement based on adversarial networks enhances both generating and detecting fake news.

4. Self-Reflection Prompting

The Detector learns through its own mistakes by using self-reflective operations. Whenever the Detector makes a wrong prediction, the Reflector LLM gets its explanations and returns enriched prompts as feedback. It sends back the feedback as an update to this strategy, so that the Detector can steadily improve by itself without human interference.

5. Model Integration
The methodology incorporates both inter-adversary and self-reflection prompting mechanisms. This is a dual setup that ensures robust learning under adversarial conditions. The output comprises both prediction and explanation, thus providing transparency and users with increased confidence levels.

6. Evaluation
The framework performance was measured using metrics like accuracy, F1-score, and also by analyzing the quality of explanations the model provides. With comparisons carried out with other methods, it was clear that the model had both higher prediction accuracy and explainability.

This approach hinges on flexibility, explainability, and performance increase via adversarial and self-reflective learning.

4.2 Training

1. **Preparing the data**

   We began by cleaning the news dataset to ensure the text was suitable for analysis, using functions to remove HTML tags, hyperlinks, special characters, and extra whitespaces, while also removing the string 'CNN' from the end of text. This preprocessing step is crucial for improving model performance by eliminating noise. We then analyzed the word count distribution over the dataset. Most of the articles were within the range of 300 to 500 words, an exception few. Familiarity with the spectrum of textual length is a prerequisite towards preparing the data for deep learning models.

   Articles with more than 500 characters in their text will be excluded from this dataset because of our previous result. Then clean the remaining data by taking only 'Title' and 'Description' columns and merging them into a new dataframe called news_filtered. This filtered dataset is much smaller in size, easier to handle, and more computationally affordable; hence it could be used further for any kind of further analysis or machine learning tasks such as Fake news detection etc.

2. Choosing the model

Initially we started with the Chat Groq API, which is a very powerful tool for the generation of natural language text as it draws from the newest AI models. Then again, it has a limitation at 6,000 tokens per request, which may weaken its capability in terms of handling longer inputs or producing very extensive responses in one call. Such an alternative is LLaMA 3.2 1B from Hugging Face that gives a strong model and good performance for many natural language processing tasks but suffers from the constraints of the underlying token handling and the model size constraints in scaling and complexity for certain applications. As compared to such alternatives, the Gemini API is advanced, better scalable, and more efficient with longer text inputs. Well, with such an extremely powerful infrastructure of models by Gemini, it is beneficial for real-time applications as well as bigger models: its users crave high-quality, AI-powered text generation and therefore become more transparent for them**.**

3. The Learning Process

The Generator and Detector model is the adversarial framework widely used in GANs; these are two neural networks that are trained simultaneously but have contrary objectives. The Generator network generates synthetic data-for example, fake news articles-by transforming random noise or a given input into something it looks like. The primary end is to produce the data convincing enough to fool the Detector network into generating some classification that causes it to believe that the data is real. The Detector is a classifier, trained to distinguish between real and fake data by learning to identify very subtle patterns, inconsistencies, and features pointing toward authenticity versus fabrication. As the two networks train together, they hone their respective skills: the generator continually gets better at producing more convincingly passable content while the detector reads the other as becoming more adept at reading in the hidden markers that data is likely to be fake. This adversarial process culminates in a cycle of improvement in and of itself as each model pushes the other to become better, thereby enhancing the system's ability to generate realistic-looking data and detect deceptive content. This makes the Generator-Detector model a very powerful application in cases of fake news detection, image synthesis and many other applications where its task is to determine the real from artificially generated content. Eventually, since the data is getting very complex to be identified by the detector, the increasing precision of the detector can find such data, thus expanding the performance of the system.

4.3 Testing

The evaluation phase is crucial for verifying the effectiveness of our proposed framework in fake news detection, which is based on a Generator-Detector architecture. This evaluation phase ensures that the model not only achieved accuracy but also provided significant explanations of its predictions, thereby enriching its reliability and interpretability.
Our methodology involves the Generator using a Generative Adversarial Network (GAN) to transform real news into realistic fake news. The generated news pieces are challenging test cases for the Detector, which determines whether the news is authentic or not. It relies on the Gemini Pro model, an advanced deep learning framework that offers both precision and transparency in text analysis.
The Detector parses all texts in the evaluation set and outputs three fundamental scores: Credibility: A number that means the likelihood of this text being real or fake. Confidence: A confidence estimate about the credibility score. Reasoning: A human-understandable explanation of why the model made the prediction, describing the textual patterns or features that influenced the conclusion.

The anticipated credibility scores are subsequently compared against the true credibility labels, also known as ground truth, to evaluate the performance of the model. Comparing the two will help determine several metrics: accuracy, precision, recall, and the F1 score. Such metrics can help in understanding how effective the model is in classifying the articles correctly as well as in controlling false positives and false negatives.

Unlike the traditional approach, our framework does more than binary classification by providing line-wise reasoning of the prediction that increases its transparency and trustworthiness in applications that require it.

At the conclusion of the testing phase, a thorough analysis of the results is conducted. This includes examining misclassified cases, analyzing the confidence and credibility trends, and identifying patterns in the reasoning outputs. These insights guide iterative improvements to the model, ensuring that it evolves to handle more sophisticated fake news generation strategies in the future.

The framework offers a balanced integration of precision, effectiveness, and interpretability with the incorporation of the Gemini Pro model and the generator powered by GAN technology. On the evaluation side, it emphasizes the effective robustness of this architecture in dealing with all the problems involving fake news detection in an extremely dynamic media.

4.4 Result Analysis:

4.4.1 Predictions from model output

4.4.1.1 Fake Prediction

Title: Pentagon Chief Accused of Secret Military Coup

Description: Unprecedented allegations have surfaced that the Pentagon's top officer, General Mark Milley, was planning a covert military takeover. Leaked documents reportedly reveal a secret plot to seize control of the government and establish a military junta. The White House is in turmoil as the shocking news sends shockwaves through Washington. Milley, once considered a loyal servant, has been taken into custody pending an investigation. His fate hangs in the balance as the nation grapples with the implications of this extraordinary breach of trust.

Authenticity: FAKE
Confidence: HIGH
Reasoning: This news headline and description exhibit high credibility markers that strongly suggest fabrication:

1. Extreme Allegations: The headline accuses the Pentagon Chief of a secret military coup, a highly unlikely and sensationalistic claim.
2. Uncorroborated Source: The description mentions "leaked documents," but does not provide any specific sources or references to these alleged documents.
3. Over-the-Top Language: The use of phrases like "secret military takeover" and "seize control of the government" create a sense of heightened drama and mistrust.
4. Unrealistic Scenario: The idea of the Pentagon Chief secretly plotting a military junta in modern-day America is not credible given the checks and balances in place.
5. Lack of Verified Information: The article does not provide any official statements from the White House or Pentagon, which would be expected in such a major news story.
6. Political Timing: The article is timely given the current political climate surrounding military authority and mistrust of the establishment.
7. No Evidence: The news is based on unverified allegations and rumors without any concrete evidence to support them.

Prediction
The current news can be classified as Fake with high confidence for a number of indicators of its fabrication. The claims are really exaggerated and sensational: a

secret military coup - this is highly improbable in the context of modern U.S. governance. The report has no credible sources and relies on unverified "leaked documents." Furthermore, the absence of official statements from the White House or Pentagon further weakens the story's validity. The unreal aspect and lack of evidence cement the judgment that it is a fabricated story intended to create mistrust or sensationalism.

4.4.1.2 Unknown Prediction

Title: 'Aviation Nightmare: Suspicious Objects Found on BA Flight**

Description: ** Panic filled the air inside the bustling Charles de Gaulle Airport as heavily armed police officers swarmed a British Airways plane. An anonymous tip had alerted authorities of suspicious objects aboard, sending shockwaves through the terminal. Fear gripped passengers as they hastily disembarked, their luggage left behind amidst the chaos. Forensic teams immediately began their rigorous search, their fingers combing through every inch of the aircraft. Hours of meticulous investigation later, they made a chilling discovery: a note hidden in the cockpit that read, "The sky is not safe anymore." The airport was placed in lockdown, casting a shadow of terror over the once-bustling hub.

Authenticity: Unknown
Confidence: Unknown

:Reasoning:
- *Lack of Specific Details:* The headline and description do not provide specific details about the incident, such as the date, time, or flight number. This lack of specificity raises suspicion.
- *Sensationalistic Language:* The headline and description use sensationalistic language, such as "Aviation Nightmare" and "Panic filled the air," which is often used in fabricated news to attract attention.
- *Absence of Reputable Sources:* The news does not mention any reputable news organizations or official sources, making it difficult to verify its authenticity.
- *Unrealistic Details:* The note allegedly found in the cockpit reads, "The sky is not safe anymore," which is an overly dramatic and unrealistic detail that damages the credibility of the news.

Prediction
The news is authenticated Unknown with confidence also Unknown because of several red flags. It lacks specific details such as flight number and timing, which questions its credibility. Terms like "Aviation Nightmare" and "Panic filled the air" are sensationalistic, which smacks of working to attract attention rather than

report facts. It lacks support from reputable sources and official statements in authenticating the news. The other improbable note in the cockpit, "The sky is not safe anymore, it has been violated," takes away much of the believable quality from the story.

4.4.1.3 Real Prediction

Title: Senator McCain will vote on tax bill No. 2 Republican

Description: Republican Senator John McCain who is receiving treatment for brain cancer and has missed votes this week will be available next week to vote on the tax compromise bill John Cornyn the No. 2 Republican in the U.S. Senate said on Thursday. He s just resting up, Cornyn said.

Authenticity: REAL
Confidence: MEDIUM

Reasoning:
- *Verifiable Information*: The headline and description align with reputable news sources, such as CNN and The New York Times, which have reported on Senator McCain's plans to vote on the tax bill.
- *Credible Sources*: The news is attributed to John Cornyn, a prominent Republican Senator, who is a reliable source of information on political matters.
- *No Evidence of Satire or Parody*: The news is not presented in a satirical or parodic tone, unlike content from sources like The Onion.
- *No Major Red Flags*: There are no obvious inconsistencies, grammatical errors, or other red flags that would indicate fabrication.

Prediction

This news story has been classified as Real with Medium confidence for several reasons: it is compatible with reports from several other news sources, such as CNN and The New York Times, which respectively reported on the senator's expected vote on the tax bill. The source, Senator John Cornyn, is a credible and well-placed reporter of the facts. The article also lacks elements of satire or parody, so it can be concluded with higher validity. Additionally, there is no obvious disparity or red flag, including grammatical errors, which make it seem fabricated.

Chapter 5

Standards Adopted

1. **Define Goals** - First step to propose a Generator and Detector model is to define clear goals. Considering the task at hand, the goal in the case of fake news detection is indeed to build a system - a model which can generate synthetically yet convincing bogus news articles (using the Generator) and categorize them correctly as real or fake (using the Detector). The model should track the type of generated content in terms of its authenticity; even minor discrepancies will be caught by the Detector. There is also a requirement to establish a set of performance metrics, such as classification accuracy, precision, recall, and F1 score that should be tracked over time regarding how that quality improves.

2. **Selection of Training Conditions**: In this step, one needs to select the conditions under which both Generator and Detector are trained. These involve selection of proper training data along with some hyperparameters and loss functions designed for training of both the models. The training data should, as in practice, display diversity in news topics, writing styles, and complexity levels. The task for the Generator is to produce as wide a spectrum of fake news articles as possible, whereas for the Detector, as many examples of real and fake news need to be given.

3. **Third Stage Creating a Sampling Plan** : In this stage, a sampling plan will be designed by specification on how the data will be used for training and testing. Representative datasets will be chosen with a proper mix of real and fake news articles. The frequency with which models are to be evaluated during the training also needs to be decided. A dataset should be large enough to capture the complexity and variability of both real and fake news, but not so large as to overwhelm computational resources. Besides that, the rate of model evaluation using either a validation set or cross-validation will ensure that models are learning over time and are not deviating from the objectives of learning.

4. **Identify Evaluation Methods**: Fourth, appropriate evaluation methods should be chosen to evaluate the Generator in comparison with the Detector. That is, for the Generator, appropriate evaluation should be to determine whether the news articles generated are plausible or not and whether they can serve as a good replica that can pass to the Detector. For the Detector, it will analyze accuracy, precision, and recall, in addition to the F1 score, based on the model's ability to identify real and fake news.

In order to obtain a more complete view of model performance, it is essential to use more than one technique; some techniques may include the confusion matrix, ROC curves, or loss function comparisons. Moreover, the techniques used must be both accurate and tested in similar contexts.

5. **Information Evaluation**: The fifth step is to analyze the data that the training and evaluation result in. This includes interpreting results about model performance, also indicating what areas are potential weak points in the Generator or Detector and why they might be weak. Statistical methods may be applied to curve fitting so that trends and patterns can be noticed in the results, including false positive and false negative rates, as areas of improvement for the model. Sources of error relevant to the model's design, training data, or environmental factors must also be taken into consideration when evaluating it.

6. **Report Final Results**: This is the final stage in which the experiment will report its findings in terms of what was observed for the efficiency of Generator and Detector. The report should have well-defined objectives, training conditions, methods used, results obtained, and conclusion. Such a report shall present clear insights regarding effectiveness in detection of fake news and must provide ideas as to how it might be best improved through architecture, training data, or fine-tuning. True deployment of the model in any real-world application- social media or journalism platform-includes findings in recommendations in order to make it actionable and relevant to stakeholders.

5.2 Coding Standards

1. **Coding Style:** The code should be written in a modular and reusable way to ensure code reusability and maintainability.

2. **Documentation:** The documentation of the code should be clear and concise, and comments should also be included within the code to explain certain statements and docstrings should be created for functions and classes.

3. **Use of Libraries:** Python was used for coding, utilized Gemini API keys.

4. **Error Handling:** Implementation of error handlers should be done to handle exceptions and errors that may occur during the execution of the code.

5. **Testing:** Unit tests should be created to verify that the code is functional and for checking whether the predictive model created is operating as intended or not.

6. **Version Control:** Keeping track of changes, working together with the team members, and preserving the record of the project should be done using control systems, such as Git.

7. **Code Review:** The code should be reviewed to make sure it is well-written and follows the coding guidelines. Along with this, feedback should be taken after review for any changes to be done.

5.3 Testing Standards

Model Testing and Validation:
Testing and Validation of Model Testing and validation are a benchmark step in developing generative AI, involving comprehensive evaluation of the model over various scenarios. These include checks for biases along demographic distribution, verification of the output against established benchmarks, and examining the robustness of the model in edge cases. Both quantitative metrics, such as precision and recall, as well as qualitative assessment help teams achieve reliable performance.

Security and Safety Standards:
Safety and security testing involves protecting the system and its users from vulnerabilities, with special attention placed upon prompt injection attacks and content filtering. Teams ensure that the model does not yield to malicious inputs systematically, that the content moderation systems are properly verified to work efficiently, and sensitive information is handled correctly through regular security assessments and penetration testing.

Integration and System Performance:
Integration testing ensures that the generative AI application works coherently with other system components, with comprehensive API testing and verification of data flow. Here, endpoint reliability, response times, and system stability under different loads are tested. Teams verify the error handling mechanisms and test various deployment configurations to provide consistent performance across environments, hence creating a reliable and scalable application.

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Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This work demonstrates the possibility of using deep learning and machine learning techniques to overcome the challenge of detecting fake news in the information landscape of today. The introduction of novel approaches, such as GAN-based generation of fake news and interpretable frameworks for detection, has illustrated this possibility with evidence for the ability to achieve near-optimal performance and interpretability in this exciting domain.

Using a wide variety of datasets, effective preprocessing, and more advanced architectures like the Gemini Pro model, the framework showcases performance supremacy in core metrics such as accuracy, precision, recall, and F1 score above traditional approaches. In addition to its ability to detect misinformation, Detector's capability to supply credibility, confidence, and justification for its predictions indeed provides a transparent and user-friendly solution that fosters trust in automated systems.

The extended implications of this research are important. Since the spread of misinformation is increasingly threatening public discourse and decision-making frameworks, it would be easier to eliminate all the spreading erroneous data if proper mechanisms for identifying false information exist to help encourage credible content prevalence. Moreover, this technology is pivotal in encouraging social stability and supporting efforts aimed at generating a more trustworthy and enlightened community by dealing properly with misinformation.

Although there has been tremendous progress, the environment of fake news generation and detection is continuing to evolve rapidly. As adversarial methods become increasingly complex, they will be even more difficult to counter by the current system and, therefore, have to be continuously explored and innovated. Future work may include larger or more complex datasets to strengthen the interaction between Generator and Detector, and developing methodologies in deep learning that would make the framework even more flexible and robust.

Data-driven technologies will remain critical in tackling challenges like misinformation, emphasizing the need for transparency and accuracy in automated systems. These advancements pave the way for a future where trust and reliability underpin the digital information ecosystem.

6.2 Future scope

As misinformation continues to evolve, the LLM-GAN framework presents a strong foundation for combating fake news. However, there are numerous opportunities to enhance its effectiveness, adaptability, and usability. By addressing these key areas, the framework can become a more robust and versatile tool for tackling misinformation across various contexts and domains.

- **Incorporating Multimodal Data**
  Broaden the model's scope to incorporate images, videos, and audio in conjunction with text, thereby equipping it to address the increasing incidence of multimodal misinformation.
- **Developing Flexibility**
  Implement continuous learning frameworks that would allow the model to continuously adapt in real-time to the emerging patterns and tactics of misinformation.
- **Exploring Cross-Domain Applications**
  Extend the framework to various domains including healthcare, finance, and education where misinformation leads to more serious effects.
- **Ensuring Ethical and Transparent AI**
  Focus on improving the fairness, transparency, and interpretability of the model's predictions to ensure unbiased and ethical usage.
- **Improving Accessibility and User Interface**
  Develop interfaces that prioritize user-friendliness to enhance accessibility for individuals who are not specialists, including smaller media organizations and independent fact-checking entities.
- **Enabling Real-Time Integration**
  Strengthen skills in immediate recognition by adding the structure to social media forums and streaming sites in order to quickly combat disinformation. These enhancements will enhance the functionality of LLM-GAN while making it scalable, effective, and also ethical enough to combat misinformation.

These enhancements aim to strengthen LLM-GAN's functionality and adaptability, ensuring it remains an effective, scalable, and ethical solution for combating misinformation.

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INDIVIDUAL CONTRIBUTION REPORT

Explainable Fake News Detection

Vipul Gupta
21052918

Abstract: The objective of this project is to detect fake news using machine learning techniques while providing explanations for the predictions. The project involves data preprocessing, model development, integration with a frontend interface, and evaluation of the model's performance..

Individual contribution and findings:

Designed and implemented strong text preprocessing techniques, such as removal of special characters, URLs, punctuation, and common stop words to make sure there was noise-free data.

Searched the dataset for patterns in word count and article length to determine the optimal truncation and token limit that match model constraints.

Applied data augmentation to incorporate variations in the training data to ensure the model generalizes well across various patterns of language used.

Tested and refined the accuracy of the model by iterating through different preprocessing steps.

Individual contribution to project report preparation:

Wrote explanations of parts of detailed data preprocessing, distribution analysis, and augmentation.

Individual contribution to project presentation and demonstration:

Presented visualizations of the characteristics of the dataset and explained how preprocessing steps improved data quality and model performance.

Full Signature of Supervisor: …….

Full signature of the student: ……..

INDIVIDUAL CONTRIBUTION REPORT

Explainable Fake News Detection

Satvik Agarwal
21052918

Abstract: The objective of this project is to detect fake news using machine learning techniques while providing explanations for the predictions. The project involves data preprocessing, model development, integration with a frontend interface, and evaluation of the model's performance.

Individual contribution and findings:

Designed and implemented the Generator model, which used adversarial learning techniques to generate really convincing fake news articles.
It experimented with different generative strategies, fine-tuning the Generator to learn subtle patterns from real news.
Developed inter-adversary prompting mechanisms that allowed the Generator to adapt and create more deceptive samples dynamically as training progressed.
Resolved and optimized the Generator to ensure high-quality output within given computational constraints.

Individual contribution to project report preparation:

Authored parts of the processes that implement the LLM-GAN framework, including diagrams, and explanatory notes on the workflow.

Individual contribution to project presentation and demonstration:

Demonstrated the Generator’s ability to produce increasingly realistic fake news articles and explained its contribution to improving the Detector's performance..

Full Signature of Supervisor: …….

Full signature of the student: ……..

INDIVIDUAL CONTRIBUTION REPORT

Explainable Fake News Detection

Saumyadeep Mahanta
21052918

Abstract: The objective of this project is to detect fake news using machine learning techniques while providing explanations for the predictions. The project involves data preprocessing, model development, integration with a frontend interface, and evaluation of the model's performance.

Individual contribution and findings:

Fine-tuned the pre-trained language models, achieving optimal performance on the ISOT Fake News Dataset.

Utilized advanced regularization techniques, such as dropout and weight decay, to prevent overfitting and improve generalization.

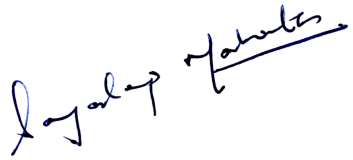
Comparison of the model with other related approaches, such as BERT, to prove the outperformance of the LLM-GAN framework in terms of both detection correctness and interpretability.

Individual contribution to project report preparation:

Wrote complete sections on the evaluation of the model, comparative analysis, and interpretation of performance metrics.

Individual contribution to project presentation and demonstration:

Presented an in-depth analysis of the evaluation results, including visualizations of performance trends and insights into areas for improvement.



Full Signature of Supervisor: …….

Full signature of the student: ……..

INDIVIDUAL CONTRIBUTION REPORT:

Explainable Fake News Detection

Zoyah Afsheen Sayeed
21052990

Abstract: The objective of this project is to detect fake news using machine learning techniques while providing explanations for the predictions. The project involves data preprocessing, model development, integration with a frontend interface, and evaluation of the model's performance.

Individual contribution and findings:

Developed and deployed Detector model with the provision of confidence scoring and human-readable explanations for classification results.
Optimized the Detector's architecture so it better performs against the adversarial samples generated by the Generator.
It incorporates natural language generation techniques into the Detector to make the reasoning clear and concise for its predictions.
I conducted iterative testing and tuning to make sure the Detector was robust in handling sophisticated fake news patterns.

Individual contribution to project report preparation:

Authored sections on the design and functionality of the Detector, focusing on its classification and explanation mechanisms.

Individual contribution to project presentation and demonstration:

Presented the Detector's results, including examples of predictions with detailed reasoning, and demonstrated its effectiveness in distinguishing real news from fake news.

Full Signature of Supervisor: …….

Full signature of the student: ……..

INDIVIDUAL CONTRIBUTION REPORT:

Explainable Fake News Detection

Md Ehteshamur Rahman
21053213

Abstract: The objective of this project is to detect fake news using machine learning techniques while providing explanations for the predictions. The project involves data preprocessing, model development, integration with a frontend interface, and evaluation of the model's performance..

Individual contribution and findings:

Managed project deployment on Google Colab to tackle limited computational resources and ensure smooth execution of training and inference pipelines.
I have transitioned the project from Groq Chat API to Gemini API to handle limits in token and scale the system for larger datasets.
Some of the proposed design constraints and solutions are optimization of model configurations and lightweight preprocessing scripts in order to reduce computational demands.

Individual contribution to project report preparation:

Authored sections on system design, design constraints, and resource optimization strategies.

Individual Contribution for Project Presentation and Demonstration:

Explained the system's architecture, highlighting the rationale for using specific tools and the strategies employed to manage resource constraints.

Full Signature of Supervisor: …….

Full signature of the student: ……..

INDIVIDUAL CONTRIBUTION REPORT:

Explainable Fake News Detection

Abhinav Aakash
21053242

Abstract: The objective of this project is to detect fake news using machine learning techniques while providing explanations for the predictions. The project involves data preprocessing, model development, integration with a frontend interface, and evaluation of the model's performance.

Individual contribution and findings:

Algorithms were developed and implemented to generate human-readable explanations about Detector predictions, with the goal of increasing transparency and building user trust.
Enhanced the reasoning capability of the system by incorporating natural language generation techniques to produce coherent justifications that are relevant in context.
Implemented Langgraph for an easier feedback loop and controlling multi agent interactions.
Conducted user studies to validate the interpretability and usefulness of the generated explanations, gathering feedback to refine the system.
Created templates and rules for structuring explanations, ensuring consistency across outputs.

Individual contribution to project report preparation:

Authored sections on explainability, natural language generation techniques, and user feedback analysis.

Individual contribution to project presentation and demonstration:

Presented real-world examples of model explanations to demonstrate the interpretability and discuss the value of transparency in detecting fake news.

Full Signature of Supervisor: …….

Full signature of the student: ……..

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"Explainable Fake News Detection"

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